Wirecard: When the System Breaks

Harvard Business School Case Study

For TUM Entrepreneurship Course: Prototyping Entrepreneurial Ideas for AI
Module 0: Startup Basics & Systems

Introduction

In February 2019, a junior venture analyst at a Munich-based VC fund received a call from a banker. Wirecard AG, once celebrated as Germany's fintech champion, was raising fresh capital to fund an ambitious expansion into AI-powered payment solutions. The timing seemed right. Wirecard's stock had just hit an all-time high of €191.80 per share. The company was valued at over €24 billion. It was the crown jewel of German technology entrepreneurship.

But there was a problem. Two days earlier, the Financial Times had published the first of what would become a damning series of investigative reports. The allegations were serious: accounting fraud, fake revenue from Asian subsidiaries, inflated transaction volumes. The company denied everything. CEO Markus Braun called the accusations "pure lies." Investors shrugged. The stock barely moved.

Our analyst had a decision to make. Her fund's partners were split. Some saw a buying opportunity. Others saw red flags. She had 48 hours to recommend whether the fund should invest in Wirecard's new AI payment subsidiary—a venture that promised to revolutionize how small businesses processed transactions.

To make this decision, she needed to understand one thing: How could a €24 billion company collapse from the inside? What did it mean when the feedback loops that kept a startup honest simply stopped working?

Background: The Wirecard Story

Wirecard was founded in 1999 by Jan Marsalek and Markus Braun, two Munich entrepreneurs in their mid-20s. The company's early mission was elegant: provide

payment processing infrastructure for the emerging e-commerce market. In its first years, Wirecard was a B2B plumbing company—it sat between online merchants and banks, taking a small percentage of each transaction.

By 2005, the company had achieved a modest milestone. It had profitable operations, steady cash flow, and a clear business model. But it was small. Wirecard was generating perhaps €30 million in annual revenue and had a valuation of maybe €100-200 million. To most of Germany's industrial elite, it was invisible.

Then Markus Braun made a crucial strategic decision: Wirecard would pursue *acquisition-driven growth*. Rather than organically expanding in Germany, the company would acquire payment processors across Europe, Asia, and the Middle East. Each acquisition would contribute revenue to the top line. Each acquisition would increase transaction volume. Each acquisition would make Wirecard's annual reports look more impressive.

The strategy worked—at least on paper.

The Acquisition Machine (2007-2018)

Between 2007 and 2015, Wirecard acquired dozens of companies. By 2010, the company had achieved its first billion-euro valuation. By 2015, it had entered the DAX 30 (Germany's blue-chip stock index). By 2018, it was valued at over €20 billion.

But here's where the system began to break.

When a young startup acquires another company, several things must happen for the deal to create value:

1. **Integration**: The acquired company's systems must be integrated with the acquirer's systems. Data must be standardized. Teams must be merged or kept separate strategically.
2. **Due diligence**: Before acquisition, investors and management must verify the target's financial claims. Are revenues real? Are they recurring? Are the customers creditworthy?
3. **Operational synergy**: After integration, the acquirer should extract efficiency gains. Eliminate duplicate functions. Consolidate suppliers. Improve margins.
4. **Feedback**: Auditors, customers, and analysts should ask tough questions. If the numbers don't make sense, they should flag them.

Wirecard's acquisition strategy broke on all four counts.

The Missing Feedback Loops

By 2016, Wirecard's CFO was Alexander Leichter. The company's audit committee included respected figures from German business. The company was audited annually by EY (Ernst & Young), one of the Big Four accounting firms.

On paper, this looked good. The feedback loops seemed intact.

But something strange happened. As Wirecard's acquisitions became more complex and geographically scattered, the feedback became harder to gather. How do you audit a payment processor in Thailand? How do you verify transaction volumes in the Middle East? How do you confirm that "merchant customers" in Singapore actually exist?

Wirecard's answer was elegant: you trust management.

In 2015, Wirecard reported revenue of €318 million and EBITDA of €66 million. The company's margins looked reasonable. Investors seemed satisfied. But a closer examination would have revealed something odd: more and more of Wirecard's revenue was coming from "third-party merchant acquiring" (TPA)—payments processed by other companies on Wirecard's behalf, then passed through Wirecard's accounts.

By 2018, third-party merchant acquiring represented over 48% of Wirecard's revenue. This was unusual. It meant Wirecard was increasingly dependent on partnerships rather than direct customer relationships. It meant the company wasn't actually processing most of the payments it claimed to process—it was just taking a cut.

Why would sophisticated investors accept this? Because the story made sense. Wirecard had positioned itself as a "financial infrastructure" company. Maybe it didn't need to process transactions directly. Maybe it just needed to be the intelligent middleman.

The narrative held together. The system—at least the narrative system—remained intact.

The Key Metric: What Gets Measured Gets Gamed

Here's where the entrepreneurial basics matter.

Every startup has a handful of metrics that matter. For a payment processor, the key metrics are:

- **Transaction volume**: How much money flows through the system?

- **Revenue per transaction**: What percentage of volume does the company capture?
- **Customer acquisition cost (CAC)**: How much does it cost to add a new merchant?
- **Lifetime value (LTV)**: How much does the company earn from a merchant over time?
- **Churn**: How many customers leave each year?

Wirecard's investors obsessed over one metric: **total transaction volume**. The company reported that it processed $360+ billion in transactions annually by 2018. This number was staggering. It made Wirecard look enormous—comparable to PayPal or Square.

The problem: much of this volume wasn't real.

In a healthy startup, investors and analysts directly verify the top metric. For Wirecard, this was nearly impossible. You can't fly to every corner of the world and audit every transaction. You have to trust the financial controls. You have to trust the auditors. You have to trust the CEO.

When the system is healthy, these trust relationships are precisely calibrated. Auditors question management. Analysts dig into methodology. Competitors watch for red flags.

But by 2018, Wirecard's system had developed a dangerous feedback loop in reverse:

Year 1: Wirecard reports 10% growth in transaction volume. Stock rises. CEO is celebrated.

Year 2: Wirecard reports 15% growth. Stock rises further. Investors expect growth to continue.

Year 3: Wirecard reports 18% growth. Stock is already priced in. Investors expect more aggressive growth to justify valuation.

Year 4: To meet expectations, Wirecard reports fraudulent growth. The system has become self-sustaining.

In a healthy system, when growth seems too good to be true, skeptical voices emerge. Analysts downgrade. Short-sellers investigate. Regulators ask questions.

But Wirecard had insulated itself from skepticism. The company's audit committee—nominally independent—consisted of longtime associates of the CEO. EY, the auditor,

had been with the company for years, creating a cozy relationship. Regulators in Germany largely trusted the company.

The feedback loops that should have caught the fraud had been broken.

The Situation: Munich, February 2019

On February 3, 2019, the Financial Times published its first investigation. The headline: “Wirecard: The €1.9bn Fraudster That Hides in Plain Sight.” The article, written by reporter Dan McCrum, alleged that:

- Wirecard had fabricated accounting entries to hide a €1.9 billion hole in its balance sheet.
- Much of the company’s reported revenue from Southeast Asian partners was fictitious.
- Wirecard had engaged in a pattern of accounting manipulation to inflate profits.

The FT report included specifics. It cited former employees. It referenced suspicious accounting entries. It questioned the legitimacy of major business partnerships in Thailand and the Philippines.

Wirecard’s response was swift and dismissive. CEO Markus Braun gave an interview to German media. “These allegations are absolutely baseless and of pure fantasy,” he said. “Our company is 100% clean.” The company’s PR team distributed talking points. The board issued a statement of confidence in management.

The stock barely moved. It closed at €190.50 on February 3—just 1% lower than the previous day.

Why didn’t investors panic?

Several factors:

1. **Denial is powerful**: A €24 billion company doesn’t collapse from a single newspaper article. Investors assumed that if something were truly wrong, German regulators would have caught it years ago.
2. **Wirecard had allies**: The company had supporters in German business and politics. It was a national champion. Skepticism felt almost unpatriotic.
3. **The short-selling context**: The FT investigation coincided with active short-selling campaigns against Wirecard. Investors assumed the newspaper was being

used as a tool by financial speculators trying to profit from a stock collapse. The credibility of the allegations became entangled with speculation.

4. **The system was insulated**: Wirecard's leadership had carefully cultivated relationships with key stakeholders. Directors trusted the CEO. Auditors had found no irregularities (because they hadn't looked hard enough). Politicians had backed the company. The system had insulated itself against external criticism.

The Analyst's Dilemma

Our junior analyst faced a specific decision. Her fund had been approached about investing in Wirecard's new subsidiary—a venture that would use artificial intelligence to detect fraudulent transactions and optimize payment routing.

The AI payment subsidiary was a real business opportunity. The technology made sense. The market opportunity was large. But the subsidiary would exist within Wirecard AG. It would depend on Wirecard's infrastructure, Wirecard's customer relationships, and Wirecard's funding.

Her fund's partners were split:

The optimists argued: "The FT allegations are speculation. Wirecard has been through audits. The company has been profitable for years. This is an opportunity to get in on the ground floor of an exciting AI venture at a discount. The market cap might recover once these allegations blow over."

The skeptics argued: "We don't know what's real and what's manufactured. The accounting doesn't quite add up. Why has the company been so successful in Asia when we can't verify anything there? Why are the auditors so passive? Something feels off."

The analyst had to decide. But she needed a framework.

Key Financial Data

Here's what the analyst knew:

Wirecard AG - Selected Financial Data (2014-2018)

Metric	2014	2015	2016	2017	2018
Revenue (€M)	301	318	356	441	582
EBITDA (€M)	37	66	81	122	179
EBITDA Margin	12.3%	20.8%	22.8%	27.7%	30.8%
Transaction Volume (€B)	82	117	167	241	360
Stock Price (Year-end)	€5.20	€22.60	€65.50	€129.50	€161.80
Market Cap (€B)	2.5	11	16	20	24

On the surface, these numbers told a compelling story. Revenue was growing at 32% annually. Profitability was accelerating. Transaction volumes were exploding. The stock was soaring.

But the analyst noticed something odd:

Wirecard's Revenue Composition (2018)

Source	Revenue (€M)	% of Total
Acquiring	276	47%
Risk Management	147	25%
Issuing	89	15%
Third-Party Acquiring	70	12%

The "Third-Party Acquiring" line was smaller than it had been in 2017 (when it was 19% of revenue). This suggested the company was moving away from the problematic business model. It was a good sign. Or was it?

The analyst didn't have a good answer.

The Decision

It was Tuesday, February 5, 2019. The analyst had two days to make her recommendation before her fund's investment committee met.

She needed to decide: **Should the fund invest in Wirecard's AI payment subsidiary?**

But more broadly, she needed to think about what the decision meant. What did it mean to invest when the fundamental claims made by a company couldn't be verified? What were the startup basics that should govern this decision?

The Questions She Should Ask:

1. **Verification and due diligence**: How can we verify Wirecard's claimed transaction volumes and revenue? What does good due diligence look like when auditors have apparently missed red flags?
2. **Feedback loops**: Which feedback loops in Wirecard's system are broken? Where should healthy skepticism come from, and why hasn't it?
3. **Metrics and gaming**: The company's key metrics are growing exponentially. But what metrics matter most, and are they real?
4. **Burn rate and runway**: This is a subsidiary. If Wirecard's main business is fraudulent, what does that mean for the venture?
5. **The system metaphor**: Wirecard is presented as a €24 billion system. But a system is only as strong as its feedback loops. Is this system healthy?
6. **Risk assessment**: What's the downside if we're wrong? What's the upside?

The analyst sat at her desk with the analyst reports, the FT articles, and the financial statements spread in front of her. Her phone buzzed. Her partner wanted her recommendation by Wednesday morning.

She opened a blank document and began to outline her thoughts.

Exhibits

Exhibit A: Wirecard Stock Price and Transaction Volume Claims (2010-2019)

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Stock Price (€) and Reported Transaction Volume (€B) - Timeline

Year | Stock Price | Trans. Volume | Growth Rate
-----|-------------|---------------|----------
2010 | €2.40       | €12B          | -
2012 | €3.80       | €24B          | 100%
2014 | €5.20       | €82B          | 21%
2015 | €22.60      | €117B         | 43%
2016 | €65.50      | €167B         | 43%
2017 | €129.50     | €241B         | 44%
2018 | €161.80     | €360B         | 49%
2019 | €191.80 (Feb 3) | €360B    | 0%
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Context: The stock price growth nearly perfectly tracked transaction volume growth. This is unusual for a software/fintech company. Typically, stock prices are driven by profitability, not raw transaction volumes.

Exhibit B: Key Excerpt from Financial Times Investigation, February 3, 2019

“Wirecard provides payment processing services for online merchants. But much of its business is based on sales recorded in ‘special purpose entities’—companies in Asian countries that are 50%-owned by Wirecard but controlled by local partners. How much money actually flows through Wirecard’s systems in those markets remains opaque. Several people familiar with the company told the FT that transaction volumes in south-east Asia are vastly exaggerated, and that what Wirecard describes as its ‘merchant acquiring’ business there is, in fact, a vehicle for laundering fictional revenues into the consolidated group accounts.”

Exhibit C: Analyst Report - Goldman Sachs, January 2019

Rating: SELL (initiated) Price Target: €133 Current Price: €191.80

“We are initiating coverage of Wirecard with a SELL rating. While the company has achieved impressive revenue growth, we have significant concerns about the quality of earnings and the sustainability of the business model. Specifically:

1. Third-party merchant acquiring (now ~50% of revenue) lacks transparency

2. EBITDA margins of 30%+ are unusually high for a payment processor (peers: 15-20%)
3. Asia operations represent 50%+ of profits but are difficult to audit
4. Related-party transactions are not fully disclosed

We believe the stock is overvalued at 1.2x sales. Fair value is closer to €130, implying downside to our price target."

Note: Goldman Sachs initiated this sell rating in January 2019—before the FT story. It was widely dismissed by other analysts.

Exhibit D: Wirecard Board of Directors (2019)

Name	Title	Background
Markus Braun	CEO	Founder, Wirecard (1999)
Jan Marsalek	COO	Founder, Wirecard (1999)
Thomas Eichelmann	CFO	20+ years at Wirecard
Wulf H. Matthias	Chair, Audit Committee	Retired banker, multiple directorates
Detlef Trefzger	Independent Director	Former executive, KPMG
Eckart Wintzen	Independent Director	Retired entrepreneur

Context: The board had strong ties to the company's founders. Independent directors had limited financial services expertise. No one with experience in payment processing or fintech audit was present.

Exhibit E: Payment Processing Peer Comparison (2018)

Company	Revenue Growth	EBITDA Margin	Transaction Volume
Wirecard	32%	30.8%	€360B
Square	26%	28%	$85B USD
PayPal	18%	16%	$560B
Adyen	22%	25%	€150B

Context: Wirecard's growth rate and margins were comparable to, or better than, global competitors. On pure metrics, the company looked healthy. This is why many investors remained convinced.

Exhibit F: Decision Memo Template

Investment Decision Memo Proposed Investment: Wirecard AI Payment Subsidiary **Date**: February 5, 2019 **Analyst**: [Your name]

Recommendation: [Invest / Pass / Further Due Diligence]

Key Assumption Check: - [ ] We can verify Wirecard's reported transaction volumes - [ ] We understand the revenue from third-party acquiring relationships - [ ] We are confident in the auditor's work (EY) - [ ] We understand why margins are higher than peers - [ ] We have a clear path to exit if the allegations prove true - [ ] The board provides adequate oversight

One-Page Rationale: [Your analysis]

Discussion Questions

1. **What went wrong in Wirecard's system?** Identify three feedback loops that should have caught the fraud earlier. Why did each loop fail?
2. **Metrics vs. Reality**: Wirecard's key metrics (revenue, EBITDA, transaction volumes) all pointed to a healthy, growing company. What metrics were missing? What should the analyst have asked about?
3. **The role of auditors**: EY audited Wirecard annually and found no issues. What does this tell us about the limits of external audits? How would you design a verification process for a company like this?
4. **Founder psychology**: Markus Braun and Jan Marsalek had built Wirecard from zero. By 2019, they owned a company worth €24 billion. How do incentives change when you've already succeeded? What pressures might they have felt?
5. **Decision under uncertainty**: The analyst had incomplete information. How do you make investment decisions when you can't verify fundamental claims? What's the right balance between skepticism and faith in established systems?

6. **Cultural context**: Wirecard was a German national champion. This may have insulated it from criticism. How does national pride or “champion company” status affect due diligence?
7. **Short-selling and credibility**: FT’s investigation coincided with short-seller campaigns. How do you separate legitimate criticism from financial manipulation?
8. **Red flags in hindsight**: If the analyst had invested, what would have been early warning signs in 2019 that something was wrong?

Epilogue (Not for Student Reading)

The analyst recommended that the fund **pass on the investment**. Her reasoning centered on verification: if the fund couldn’t verify Wirecard’s core claims, it shouldn’t depend on Wirecard’s infrastructure.

She was right.

In June 2019, Wirecard suspended its CFO, Alexander Leichter. In July, the company admitted that €1.9 billion in cash on its balance sheet likely didn’t exist. The stock collapsed 97% in three months. By December 2019, the company had filed for insolvency.

Markus Braun and Jan Marsalek were criminally charged with fraud and accounting manipulation. Braun was convicted in November 2021 and sentenced to 8.5 years in prison.

The case became one of the largest accounting scandals in German history. It revealed the importance of the entrepreneurial basics: verify your metrics, maintain healthy feedback loops, and never let a single narrative become the truth.